

The pH Ruler

Working out what the pH scale actually measures, and why one step along it is never a small step.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/ph-ruler/

1 Before you open anything

Answer from what you already know. You are not expected to be right — the point is to have something on paper to compare against.

PREDICTION

Lemon juice sits near pH 2. Black coffee sits near pH 5. How many times more concentrated is the hydronium ion, H_3O^+ , in the lemon juice?

Write one sentence saying how you got that number.

2 Read the four numbers

Open the demonstration. The slider sets the solution; the four boxes underneath report it four different ways.

1. Drag the slider until the pH box reads **3.00**.
2. Copy all four values into the first row of the table.
3. Repeat for each pH in the left-hand column.

pH	$[\text{H}_3\text{O}^+]$ (M)	$[\text{OH}^-]$ (M)	pOH	pH + pOH
3.00				
7.00				
10.00				
13.00				

a. One column has the same value in every row. Which one, and what is that value?

b. Multiply $[\text{H}_3\text{O}^+]$ by $[\text{OH}^-]$ for any row. Do it for a second row. What do you get both times, and what is that quantity called?

Look at the label the demonstration puts next to that product. It names a temperature. Every relationship on this page holds at that temperature and nowhere else — at 50 °C neutral water sits at pH 6.63 and is still perfectly neutral.

3 What one step on the scale costs

This is the part that decides whether the pH scale makes sense to you.

1. Set the slider to **pH 4.00** and record $[\text{H}_3\text{O}^+]$.
2. Move it to **pH 5.00** — one single step — and record $[\text{H}_3\text{O}^+]$ again.
3. Divide the first concentration by the second.

Change	$[\text{H}_3\text{O}^+]$ before	$[\text{H}_3\text{O}^+]$ after	Ratio (before ÷ after)
pH 4 → 5			
pH 4 → 6			
pH 4 → 7			

a. Complete the rule: moving n units along the pH scale changes the hydronium concentration by a factor of _____.

b. Go back to your prediction on page 1. Was it right? If not, say specifically what you assumed that turned out not to be true.

c. A lake at pH 6.5 is acidified to pH 5.5 by acid rain. A newspaper reports this as “a 15% increase in acidity.” Using your rule, explain in one or two sentences why that description is badly wrong.

4 Working backwards

Every one of the four boxes can be typed into, not just read. Whichever one you type into, the other three recalculate. That is the same skill a test question asks for.

1. Click into the $[\text{H}_3\text{O}^+]$ box and type **4.2e-5**. Record what the other three become.
2. Now type **2.5e-3** into the $[\text{OH}^-]$ box instead.
3. Now type **11.30** into the **pOH** box.

You typed	$[\text{H}_3\text{O}^+]$	$[\text{OH}^-]$	pH	pOH	Acidic, basic or neutral?
$[\text{H}_3\text{O}^+] = 4.2 \times 10^{-5}$					
$[\text{OH}^-] = 2.5 \times 10^{-3}$					
pOH = 11.30					

a. For the middle row you were given a *hydroxide* concentration and the solution turned out to be basic. Explain how you could have known it was basic before doing any arithmetic.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself — and if you were wrong, write down where the arithmetic went off.

1. A solution has $[\text{H}_3\text{O}^+] = 6.8 \times 10^{-9} \text{ M}$. Find pH, pOH and $[\text{OH}^-]$, and state whether it is acidic or basic.

2. A solution has pOH = 2.40. Find $[\text{OH}^-]$, pH and $[\text{H}_3\text{O}^+]$.

3. Solution A is pH 3.20. Solution B is pH 6.70. How many times more concentrated is H_3O^+ in A than in B? Give the answer to two significant figures.

4. Explain why a pH of 7.00 is described as neutral, in terms of the two ion concentrations rather than by quoting the number 7. Then state the one condition that has to hold for 7.00 to be the neutral point at all.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Work through the real-substance presets and put them in order from most to least acidic before you check.
- Push the slider to both extremes. At pH 0 and pH 14, is either ion concentration ever actually zero? What does that tell you about the two ions coexisting?

- Find the pH at which the two ion concentrations differ by exactly a factor of 100. There are two answers — find both and explain why.